

LONGREACH, Australia

WMO: 943460

Lat: 23.4398S

Lon: 144.2830E

Elev: 632

StdP: 14.36

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	39.7	42.4	7.4	8.2	69.9	12.3	10.5	68.4	21.0	69.2	19.7	67.9	5.6	140	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	24.0	105.4	68.6	103.1	68.4	100.9	68.3	76.9	87.0	75.7	86.6	74.7	86.4	10.5	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.8	133.4	79.6	73.5	127.6	79.0	72.0	121.2	79.0	40.9	86.4	39.8	86.6	38.8	85.9	95.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.7	19.8	18.1	DB	33.8	109.0	1.7	3.3	32.6	111.4	31.6	113.3	30.6	115.2	29.4	117.5
			WB	29.8	80.3	2.3	3.7	28.1	83.0	26.8	85.2	25.5	87.2	23.8	89.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	74.9	85.9	84.3	82.0	75.5	67.3	61.3	60.6	63.6	71.6	78.5	82.6	85.9	(6)
	DBStd	10.77	4.77	4.71	4.41	4.92	5.66	5.91	5.57	5.64	6.20	5.97	5.35	5.32	(7)
	HDD50	2	0	0	0	0	0	1	1	0	0	0	0	0	(8)
	HDD65	457	0	0	0	2	41	143	160	95	14	2	0	0	(9)
	CDD50	9079	1112	960	993	765	538	339	329	420	649	882	979	1113	(10)
	CDD65	4059	647	540	528	317	113	31	23	51	213	419	529	648	(11)
	CDH74	53509	8997	7058	6538	3497	1241	351	353	916	2841	5503	7073	9141	(12)
	CDH80	29585	5428	4078	3582	1603	330	44	49	253	1340	3067	4159	5652	(13)
(14) Wind	WSAvg	9.6	10.3	10.1	9.5	9.0	8.5	8.6	8.8	9.2	9.7	10.6	10.8	10.6	(14)
(15) Precipitation	PrecAvg	16.7	2.9	3.0	2.0	1.3	0.9	0.7	0.7	0.4	0.5	0.8	1.2	2.2	(15)
	PrecMax	34.4	16.5	9.6	16.3	12.4	4.3	6.7	4.4	2.8	5.2	2.8	7.2	7.6	(16)
	PrecMin	4.2	0.1	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	(17)
	PrecStd	6.7	3.0	2.4	2.6	2.3	1.2	1.2	1.1	0.6	0.9	0.8	1.7	1.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	107.6	106.8	103.1	96.5	90.1	84.1	84.8	91.4	99.5	104.5	106.4	108.1	(19)
		MCWB	70.0	70.0	69.1	68.0	63.7	62.3	62.2	60.9	63.0	66.1	67.0	68.8	(20)
	2%	DB	104.1	102.6	99.9	93.4	86.1	81.0	80.7	86.0	94.8	100.0	102.8	105.3	(21)
		MCWB	69.8	70.1	68.8	66.0	62.8	61.5	59.3	60.0	61.5	65.1	66.6	68.2	(22)
	5%	DB	101.5	99.7	97.0	90.9	83.7	78.5	78.0	82.4	91.2	96.9	100.0	102.6	(23)
		MCWB	69.9	70.2	68.6	65.0	61.9	60.1	57.8	58.1	61.0	63.9	66.0	68.1	(24)
	10%	DB	98.8	96.9	94.2	88.6	81.0	75.2	75.1	79.3	88.0	93.8	97.0	99.7	(25)
		MCWB	70.2	70.2	68.4	64.3	60.8	58.3	56.3	56.6	60.5	63.2	65.8	68.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.8	78.3	76.9	73.4	69.3	66.1	67.4	65.5	70.3	71.8	75.0	76.4	(27)
		MCDB	88.6	88.2	84.0	83.7	80.1	77.8	74.0	80.2	81.7	88.2	86.3	88.6	(28)
	2%	WB	77.0	76.9	75.2	71.4	67.2	64.1	64.0	62.9	67.5	69.8	72.8	75.0	(29)
		MCDB	87.1	87.2	84.5	83.1	78.3	75.2	74.6	78.4	80.4	86.1	84.8	87.6	(30)
	5%	WB	75.6	75.6	73.8	69.6	65.4	62.5	61.1	60.8	65.2	68.3	71.4	73.8	(31)
		MCDB	86.6	86.8	85.0	83.1	78.0	73.2	72.8	76.8	80.4	85.7	85.4	87.0	(32)
	10%	WB	74.6	74.7	72.5	67.9	63.5	60.5	58.5	58.8	63.2	66.9	70.0	72.8	(33)
		MCDB	86.7	86.3	85.1	81.5	77.1	71.5	72.0	75.8	81.7	85.1	85.7	87.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	22.6	22.4	23.3	25.1	26.1	25.9	27.8	30.0	29.7	28.1	25.6	24.0	(35)
		MCDBR	26.6	26.3	26.5	27.0	27.6	27.7	30.1	33.2	33.1	31.1	29.0	28.0	(36)
	5% WB	MCWBR	9.4	8.3	9.6	10.9	12.4	13.1	14.9	16.1	14.3	13.0	10.9	10.8	(37)
		MCWBR	20.3	19.6	19.6	21.3	21.6	21.8	24.1	27.8	25.5	24.8	22.7	21.5	(38)
(40) Clear-Sky Solar Irradiance	taub		0.382	0.372	0.355	0.341	0.325	0.307	0.300	0.318	0.360	0.377	0.389	0.390	(40)
		taud	2.474	2.500	2.528	2.522	2.507	2.562	2.560	2.484	2.380	2.360	2.375	2.410	(41)
	Ebn at Noon		305	304	300	291	283	283	290	294	294	299	301	303	(42)
		Edh at Noon	38	36	33	31	29	26	27	31	38	41	41	40	(43)
(44) All-Sky Solar Radiation	RadAvg		2308	2240	2036	1757	1437	1262	1395	1710	2016	2304	2385	2376	(44)
	RadStd		162	136	139	130	71	82	82	80	148	148	140	155	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page